

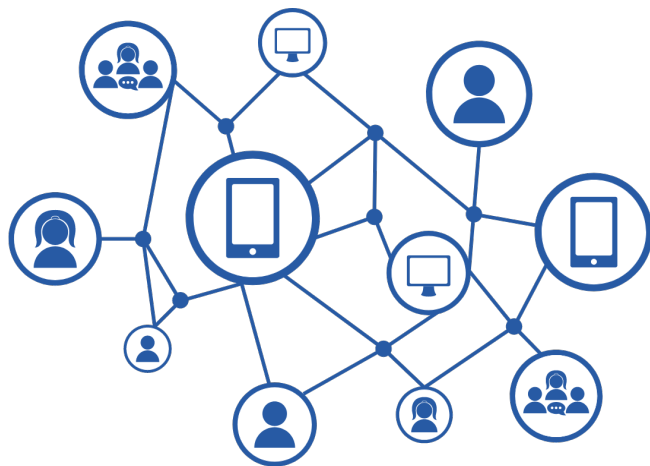
GNN's FAME: Fairness-Aware MESSAGES for Graph Neural Networks

 Erasmus Purificato, Hannan J. Mahadik, Ludovico Boratto, Ernesto W. De Luca

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Graph Neural Networks

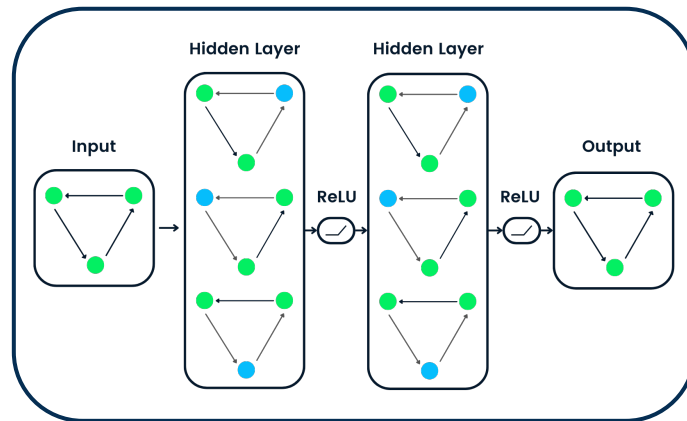
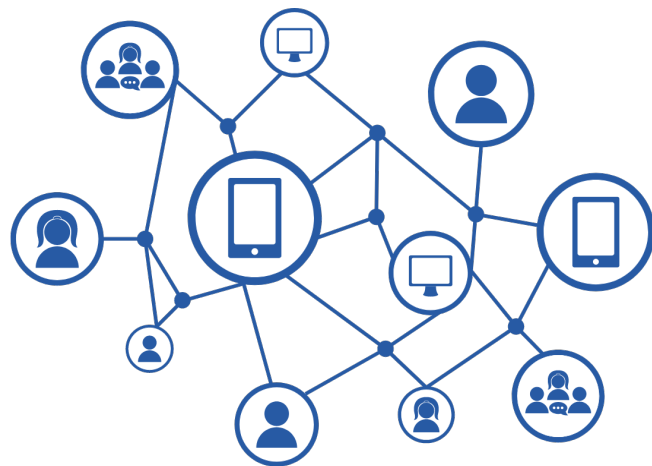
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Graph Neural Networks

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Graph Neural Networks (GNNs) represent the perfect solution.



Unfairness in GNNs

- **Inherited bias**

GNNs acquire biases from historical data

- **Amplification**

Topology worsens bias

- **Homophily effect**

Similar nodes connect more frequently



Three doors into the pipeline



Pre-processing
(Data level)



In-processing
(Learning phase)



Post-processing
(Decision level)





In-processing bias mitigation

“Inject fairness while training”



- Try to enforce algorithmic fairness by **changing the learning strategy** of the model.
- Add a **regularisation term** or another **constraint** to the existing **optimisation function**, while leaving the training data as-is.

Research questions

- RQ1: How does direct intervention in the message-passing process influence fairness metrics outcomes?

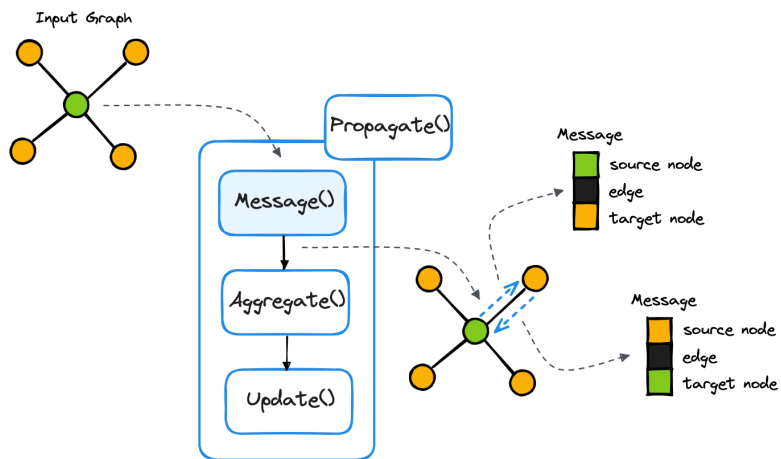


Research questions

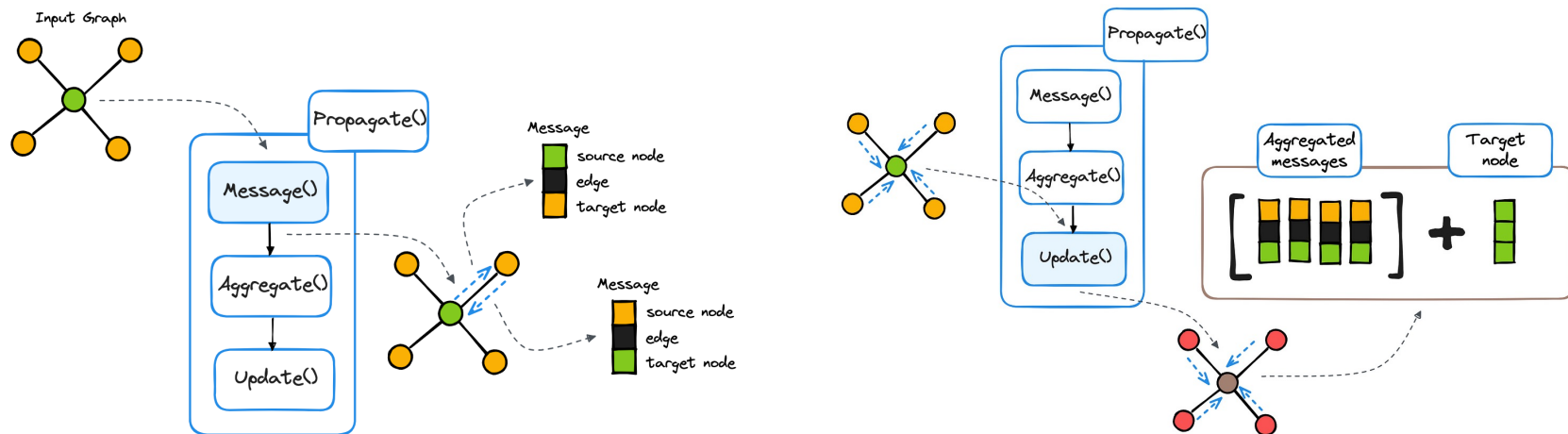
- RQ1: How does direct intervention in the message-passing process influence fairness metrics outcomes?
- RQ2: How does altering message passing affect fairness compared to state-of-the-art bias mitigation approaches?



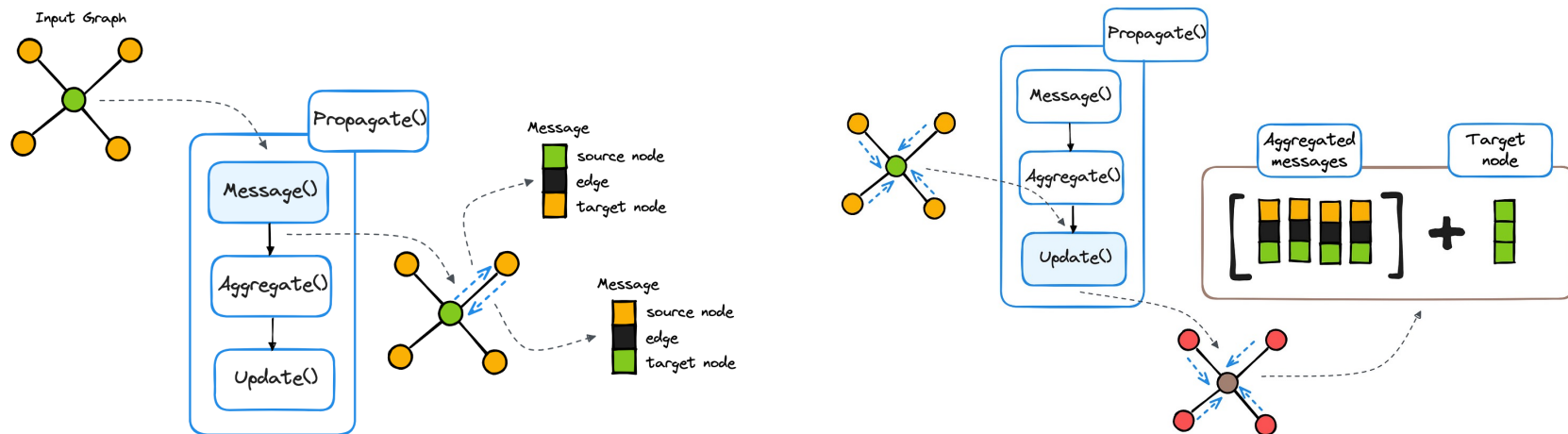
Message-passing mechanism



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Message-passing mechanism



$$\mathbf{h}_u^{(k)} = \sigma \left(\mathbf{W}^{(k)} \mathbf{m}_u^{(k)} + \mathbf{b}^{(k)} \right)$$

Fairness-Aware MESSAGES (FAME)

GCN-based models

$$\mathbf{m}_u^{(k)} = \sum_{v \in N_u \cup \{u\}} \frac{1}{\sqrt{|N_u| |N_v|}} \mathbf{h}_v^{(k-1)}$$



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$$\boldsymbol{\delta}_{uv}^{(k)} = 1 + \mathbf{b}^{(k)} \boldsymbol{\Delta}_s^{(k)}$$

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$$\alpha_{uv}^{(k)} = \text{softmax} \left(\mathbf{e}_{uv}^{(k)} + \boldsymbol{\delta}_{uv}^{(k)} \right)$$

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Evaluation

- Disparity

$$\Delta_{SP} = |P(\hat{y} = 1 \mid s = 0) - P(\hat{y} = 1 \mid s = 1)|$$

- Inequality

$$\Delta_{EO} = |P(\hat{y} = 1 \mid y = 1, s = 0) - P(\hat{y} = 1 \mid y = 1, s = 1)|$$



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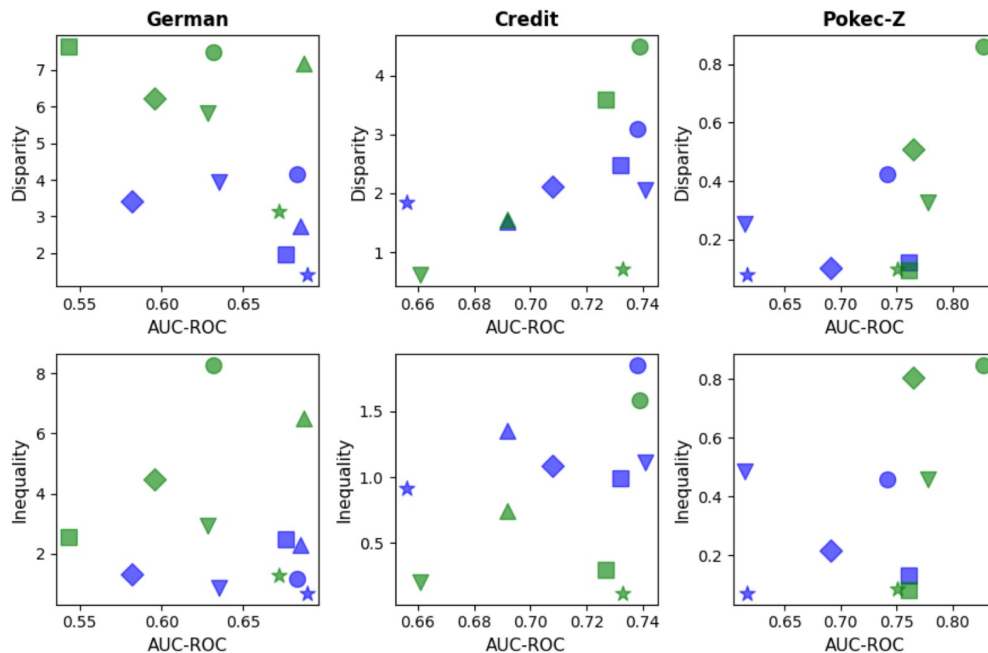
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Dataset	Nodes	Edges	Label	Sens. Attr.
German	1K	21K	good-customer	gender
Credit	30K	1.42M	no-default	age
Pokec-z	67K	617K	work-field	region

Experimental results

■ GCN ■ GAT ● Vanilla ■ Unaware ▲ EDITS ▼ PFR-AX ◆ FairGNN ★ FAME



Key observations



Effective bias mitigation

Correction term successfully reduces bias propagation



Competitive performance

Maintains accuracy while improving fairness



Promising outperformance

Better results than the selected GNN-specific baselines





Contacts and resources

Erasmus Purificato, PhD

Scientific Project Officer

European Commission, Joint Research Centre (JRC)

European Centre for Algorithmic Transparency (ECAT)



erasmo.purificato@acm.org



<https://erasmopurif.com/>

